COURSE OUTLINE



DATA9302

IBM Predictive Analytics Modeler

Fall 2021 - Current

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DATA9302 IBM Predictive Analytics Modeler

COURSE DESCRIPTION

The Predictive Analytics Modeler course provides the essential analytics tools to collect and analyze data efficiently in order to predict problems and improve outcomes. Specific skills include data mining and modeling (CRISP-DM), storage and field measurement, data collection and integration, data cleaning, nodes and streams, and statistical analysis.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	36
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Explain data analytics and its types, its importance, and how it's transforming the world.	Communication, Thinking Skills, Working with Others.
2	Outline how analytics provides solution to industries' problems using case studies.	Thinking Skills, Communication, and Digital Literacy
3	Apply data analytics to case studies.	Thinking Skills, Communication and Digital Literacy
4	Explain data transformations, and it's functions.	Thinking Skills, Communication, and Digital Literacy
5	Explain data modeling, relationships, as well as derive and reclassify fields.	Thinking Skills, Communication, and Digital Literacy
6	Explain the principles of data mining.	Thinking Skills, Communication, and Digital Literacy
7	Build predictive models	Thinking Skills, Communication, and Digital Literacy

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Orientation – Welcome Module 3 – Section 1 – Unit 1 Introduction to Data Science • List applications of data science • Explain the stages of CRISP-DM • Describe skills needed for data science
Week 2	Module 3 – Section 1 – Unit 2 Introduction to IBM SPSS Modeler • Describe SPSS Modeler's user interface • Work with nodes and streams • Generate nodes form output • Use supernodes • Execute and save streams
Week 3	Module 3 – Section 1 – Unit 3 Introduction to Data Science Using IBM SPSS Modeler • Explain the basic framework of a data-science project
Week 4	Module 3 – Section 1 – Unit 3 Introduction to Data Science Using IBM SPSS Modeler • Build a model • Deploy a model
Week 5	Module 3 – Section 1 – Unit 4 Collecting Initial Data • Explain the concepts "data structure", "unit of analysis", "field storage" • and "field measurement level" • Import Microsoft Excel files • Import IBM SPSS Statistics files • Import text files • Import from databases • Export data to various formats
Week 6	Module 3 – Section 1 – Unit 5 Understanding The Data • Audit the data • Check for invalid values • Take action for invalid values • Define blanks
Week 7	Module 3 – Section 1 – Unit 6 Setting The Unit of Analysis • Remove duplicate records • Aggregate records • Expand a categorical field into a series of flag fields • Transpose data
Week 8	Module 3 – Section 1 – Unit 7 Integrating Data • Append records from multiple datasets • Merge fields from multiple datasets • Sample records
Week 9	Module 3 – Section 1 – Unit 8 Deriving and Reclassifying Fields • Use the Control Language for Expression Manipulation (CLEM) • Derive new fields • Reclassify field values Module 3 – Section 1 – Unit 9 Identifying Relationships • Examine the relationship between two categorical fields • Examine the relationship between a categorical field and a continuous • field Examine the relationship between two continuous fields
Week 10	Module 3 – Section 1 – Unit 10 Introduction to Modeling • List three types of models • Use a supervised model • Use a segmentation model Module 3 – Section 2 – Unit 1 Using Functions to Cleanse and Enrich Data • Use date functions • Use conversion functions • Use string functions • Use statistical functions • Use missing value functions
Week 11	Module 3 – Section 2 – Unit 2 Using Additional Transformerations • Replace values with the Filler node
Week 12	Module 3 – Section 2 – Unit 3 Working With Sequence Data • Use sequence functions • Count an event across records • Expand a continuous field into a series of continuous fields with the • Restructure node • Use geospatial and time data with the Space-Time-Boxes node
Week 13	Module 3 – Section 2 – Unit 4 Sampling, Partitioning and Balancing Data • Draw simple and complex samples with the Sample node • Create a training set and testing set with the Partition node • Reduce or boost the number of records with the Balance node
Week 14	Module 3 – Section 2 – Unit 5 Improving Efficiency • Use database scalability by SQL pushback • Process outliers and missing values with the Data Audit node • Use the Set Globals node • Use parameters • Use looping and conditional execution
Week 15	Exam Preparation Week – IBM Predictive Analytics Modeler – Explorer Exam

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1-7	Foundation Badge – Minimum 60% required to pass	100%
1-7	Intermediate Badge – Minimum 60% required to pass (closed book)	0%
	Total	100%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

To pass this course learners need to obtain a grade of at least 60% on their Explorer Exam.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	Description	Value in GPA
P	Pass	N/A
F	Fail	N/A

REQUIRED LEARNING RESOURCES

Portal link: https://keyskill-clms.mylearnerportal.com/

D2L Platform

SPSS Modeler Virtual Machine

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.